



KCED

Learning Augmented

24ADI003-MACHINE LEARNING

UNIT 2: **REGRESSION MODELS**

Information Criteria–Based Model Selection

Why Do We Need Model Selection?

- In regression, we often have **many possible models**:
- Simple linear regression
- Multilinear regression
- Polynomial regression (degree 2, 3, 5, ...)
- A model with **more parameters always fits training data better**.

But:

- Better training fit does NOT guarantee better prediction
- Complex models may **overfit**
- So we need a **principled way** to choose:
 - **“A model that fits well but is not unnecessarily complex.”**
- **Information criteria** follow this rule:
 - **Good model = Good fit + Low complexity**
- They **penalize** models for having too many parameters.

Information Criteria–Based Model Selection

Error vs Likelihood

Error Value	Meaning	Likelihood
Large error	Poor prediction	Very low
Small error	Good prediction	Very high

Likelihood converts error into probability.

Assumption Used in Regression

Regression models assume that errors follow a **Gaussian (Normal) distribution**:

$$\epsilon = y - \hat{y} \sim \mathcal{N}(0, \sigma^2)$$

This assumption is standard in:

- Linear regression
- Polynomial regression

Information Criteria–Based Model Selection

Likelihood for One Data Point

Likelihood = How likely (believable) a prediction is, assuming our model is correct

- For **one data point**, we ask:
- *“If my model is correct, how likely is it to produce this actual value?”*
- Input: x_i
- Actual output: y_i
- Model prediction: $\hat{y}_i$
- The **error** is:

$$\epsilon_i = y_i - \hat{y}_i$$

- We assume:

$$\epsilon_i \sim \mathcal{N}(\mathbf{0}, \sigma^2)$$

Meaning:

- Most errors are close to **0**
- Large errors are **rare**
- Errors follow a **bell-shaped curve**

Information Criteria-Based Model Selection

Likelihood for One Data Point Likelihood Formula

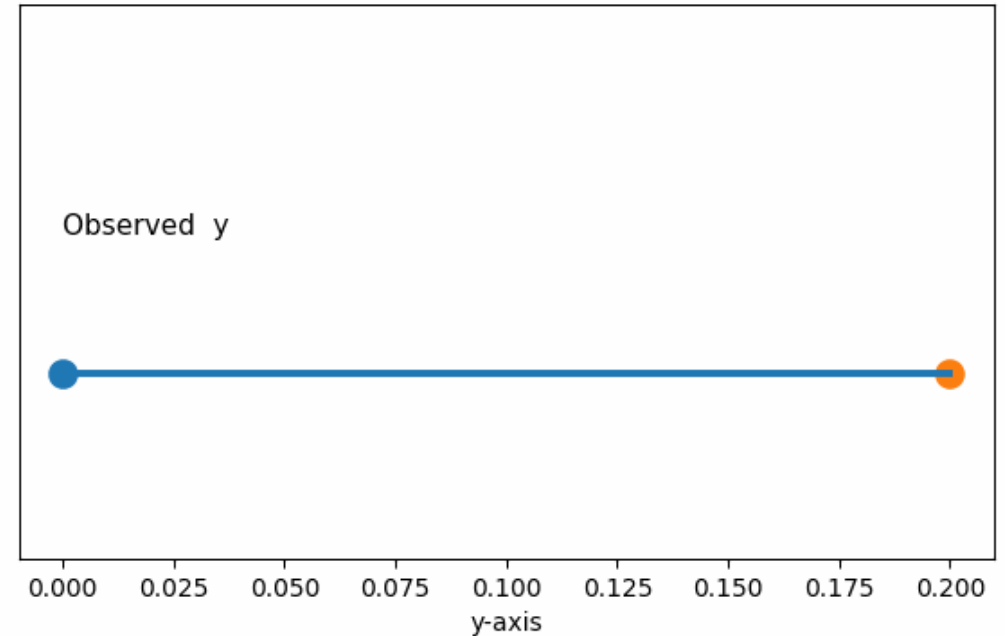
$$P(y_i|x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

Error Size	Likelihood
Very small	Very high
Medium	Medium
Very large	Very low

Example: Actual value = **50**

Prediction	Error	Likelihood
49.8	0.2	Very high
45	5	Low
30	20	Almost zero

Model vs Reality



Closer prediction → higher likelihood

Likelihood measures how good the prediction is for a single observation.

Information Criteria–Based Model Selection

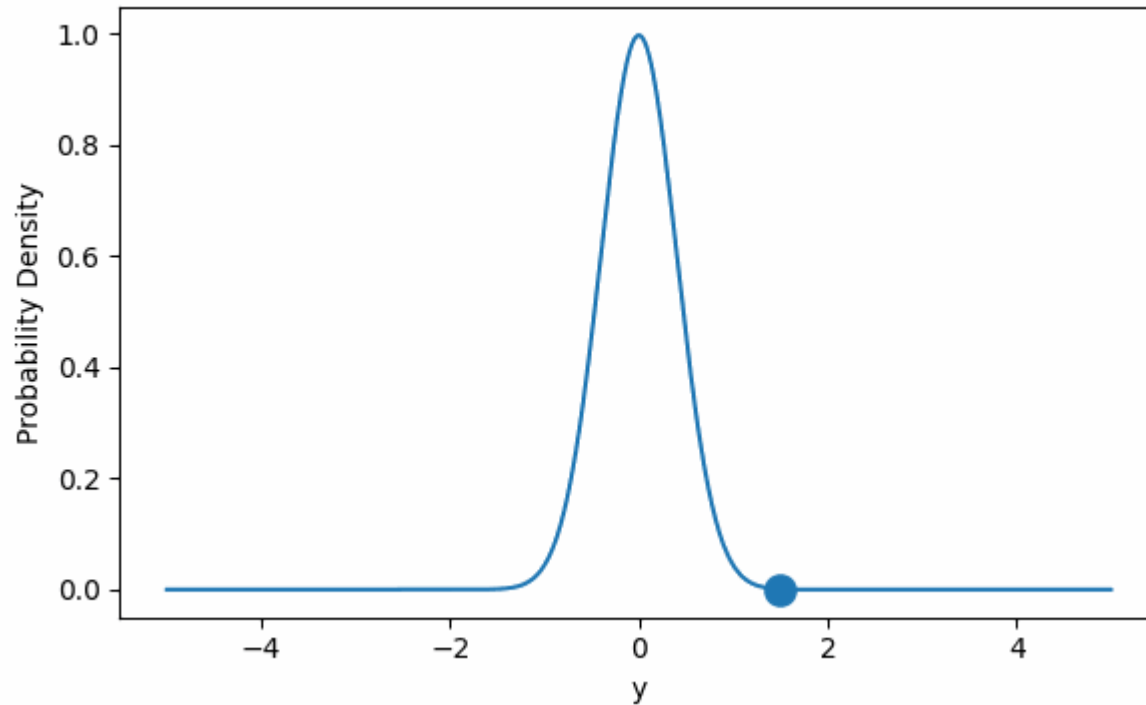
Same prediction error

↓

Narrow curve → low tolerance

$$P(y_i|x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

$\sigma = 0.40$ (Model confidence)



- Small σ → **over-confident model**
- Large σ → **uncertain model**
- Same error → different likelihoods

Same prediction error

↓

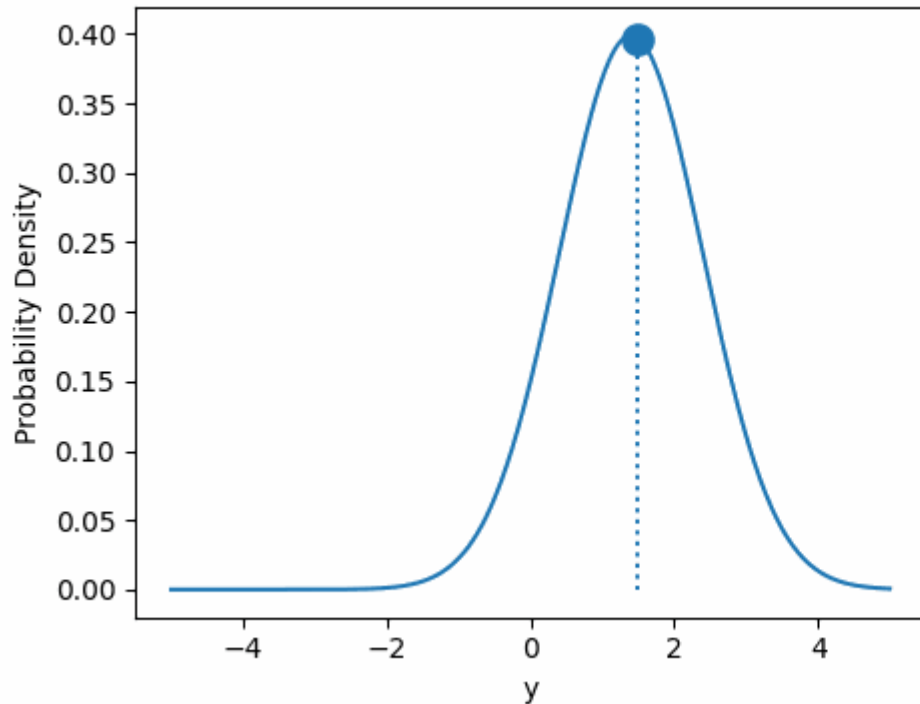
Narrow curve → low tolerance
Wide curve → high tolerance



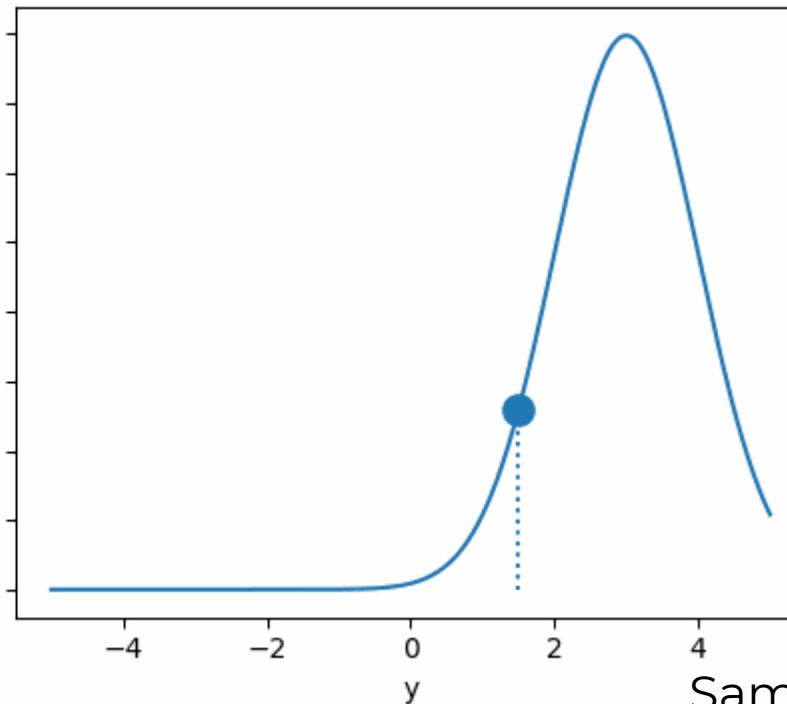
Information Criteria-Based Model Selection

$$P(y_i|x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

Good Model



Bad Model



Same σ

↓

Small error → tall likelihood

Large error → short likelihood

- Better prediction → higher likelihood
- Likelihood measures **model quality**

Information Criteria–Based Model Selection

Likelihood for Entire Dataset

A model is **not judged by one point**, but by **all data points together**.

Suppose we have:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Independence Assumption

We assume:

Errors of different points are **independent**

So probabilities **multiply**.

Likelihood of the Dataset

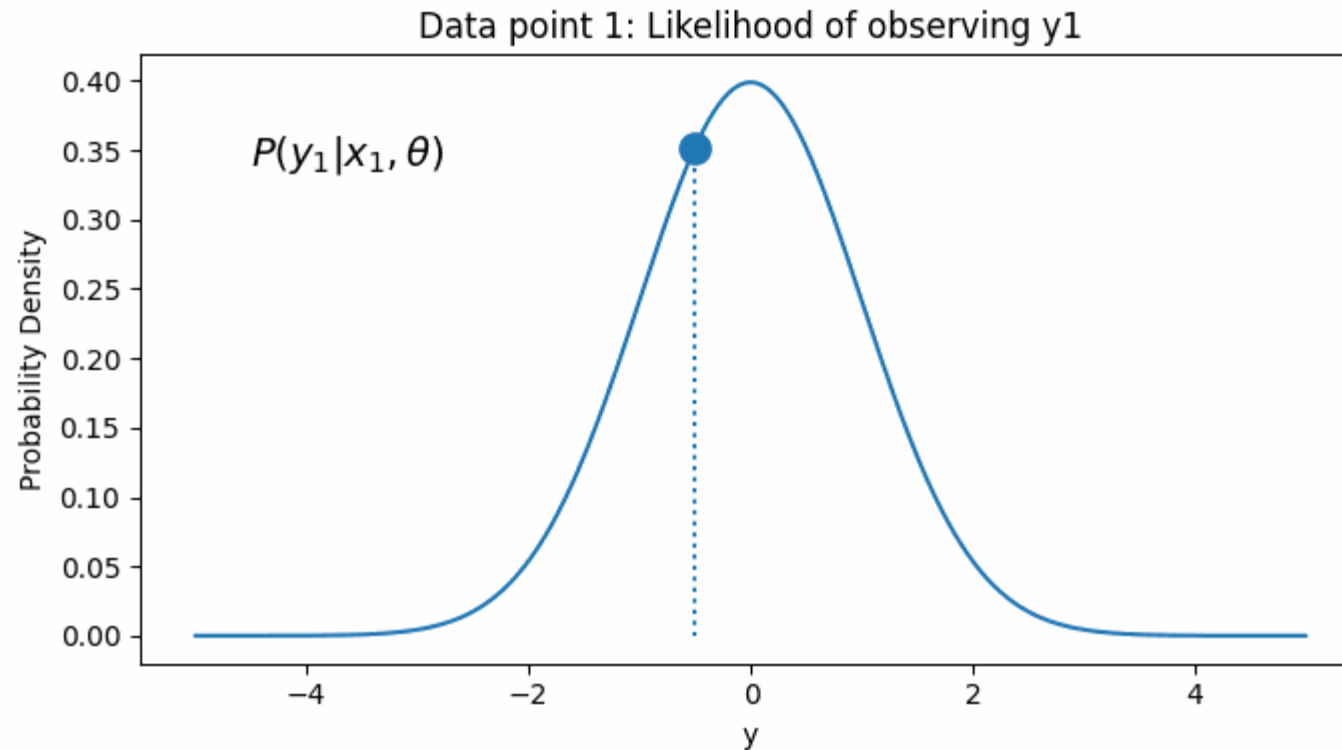
$$L(\theta) = \prod_{i=1}^n P(y_i | x_i, \theta)$$

- If **even one point** has very low probability,
- The **entire likelihood becomes small**

A good model must explain **all points reasonably well**.

Information Criteria-Based Model Selection

Likelihood for Entire Dataset



Overall likelihood measures how well the model explains all data points together.

Information Criteria–Based Model Selection

Likelihood for Entire Dataset

Example

Data Point	Likelihood
Point 1	0.9
Point 2	0.8
Point 3	0.95

$$L=0.9 \times 0.8 \times 0.95 = 0.684$$

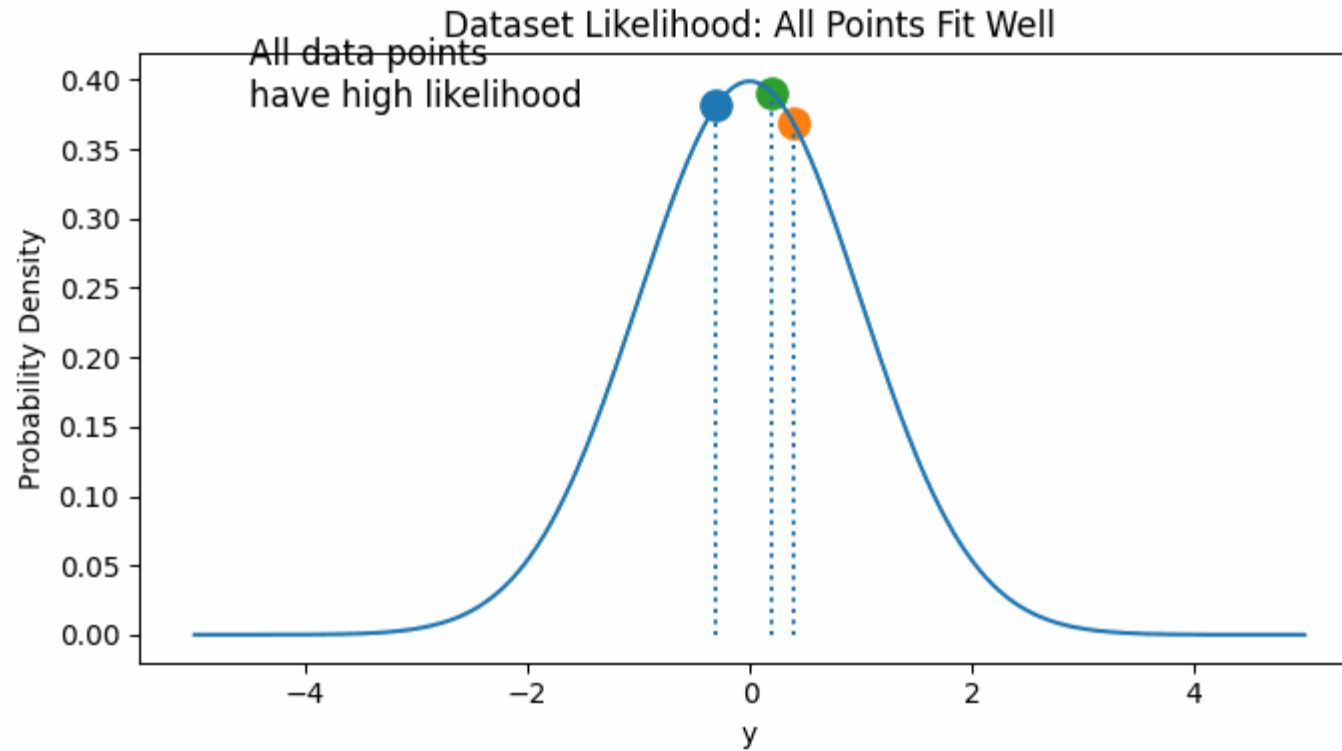
If one point is bad:

Data Point	Likelihood
Point 1	0.9
Point 2	0.01
Point 3	0.95

$$L=0.00855(\text{very small})$$

Information Criteria-Based Model Selection

Likelihood for Entire Dataset



Information Criteria-Based Model Selection

Log-Likelihood Expression

Why do we take logarithm?

Multiplying many small numbers:

Causes numerical problems

Is hard to differentiate

Logarithm helps because:

$$\log(a \times b) = \log a + \log b$$

Apply log to likelihood

$$\ln(L) = \sum_{i=1}^n \ln P(y_i | x_i, \theta)$$

After substitution and simplification

$$\ln(L) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Information Criteria-Based Model Selection

Log-Likelihood Expression

$$\ln(L) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$-\frac{n}{2} \ln(2\pi\sigma^2)$$

- Does NOT depend on model parameters
- Can be ignored during optimization

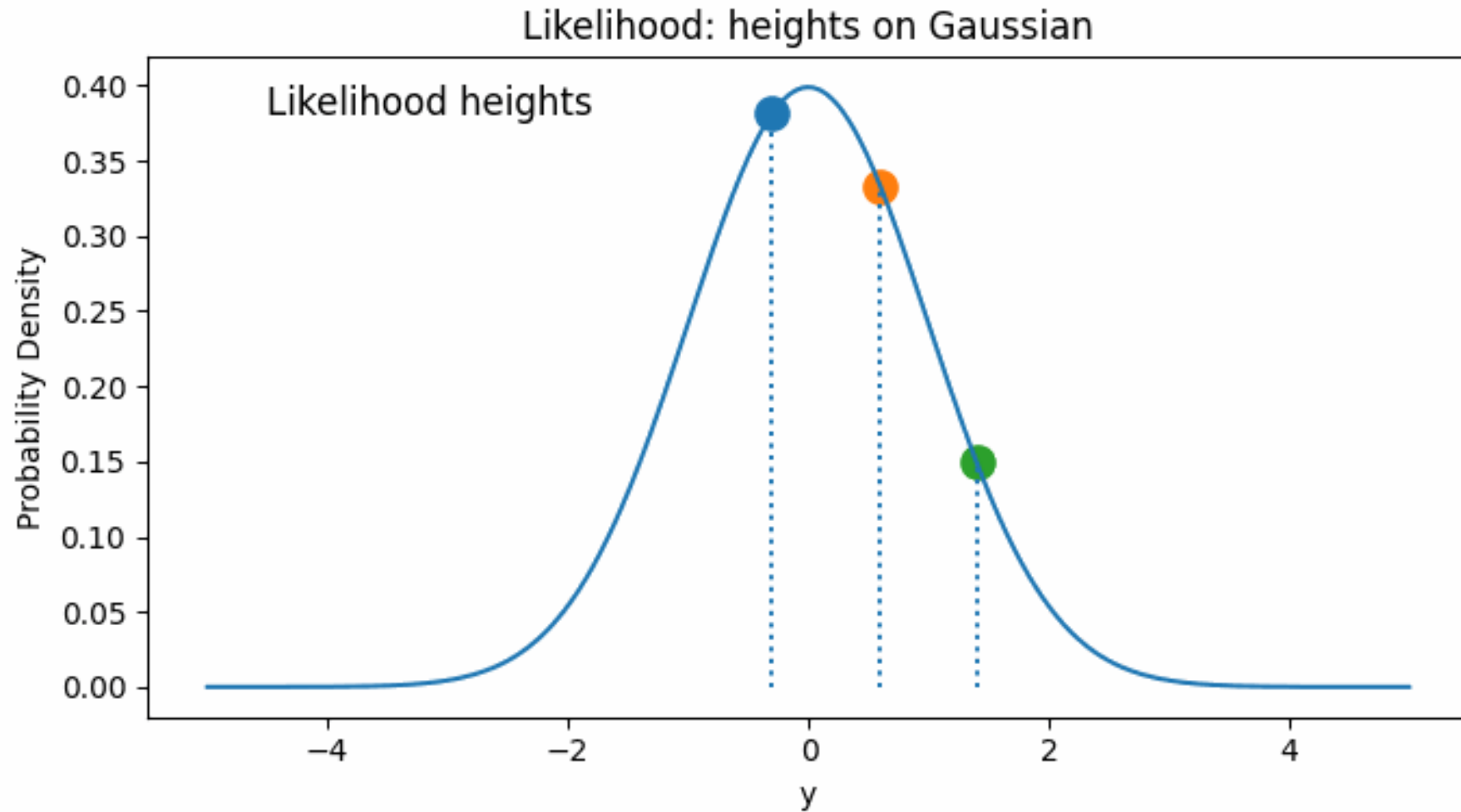
$$-\frac{1}{2\sigma^2} \sum (y_i - \hat{y}_i)^2$$

- **Depends on prediction errors**
- **Controls model quality**

Maximizing log-likelihood means:

$$\text{Maximize } \ln(L) \Leftrightarrow \text{Minimize } \sum (y_i - \hat{y}_i)^2$$

Information Criteria-Based Model Selection



Information Criteria-Based Model Selection

Likelihood for One Data Point

$$P(y_i|x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

Likelihood for Entire Dataset

$$L(\theta) = \prod_{i=1}^n P(y_i|x_i, \theta)$$

Since products are hard to compute, we use **log-likelihood**.

Log-Likelihood Expression

$$\ln(L) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \hat{y}_i)^2$$

$$\ln(L) \propto -\sum (y_i - \hat{y}_i)^2$$

Minimizing squared error = Maximizing likelihood

Information Criteria–Based Model Selection

Akaike Information Criterion (AIC)

- Hirotugu Akaike noticed a **fundamental paradox** in model building:
- A model that fits training data extremely well may perform very badly on new, unseen data.

Why does this happen?

- Because the model:
- Learns **noise**, not just the signal
- Becomes **too complex**
- Loses **generalization ability**
- This problem is known as **overfitting**.
- *How well will this model predict new data drawn from the same process?*
- AIC does **not** try to find the “true model”.
Instead, it tries to find the **best predictive model**.

Information Criteria–Based Model Selection

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Information Criteria–Based Model Selection

Akaike Information Criterion (AIC)

Core Philosophy of AIC

- AIC is based on **information theory**.
- It estimates:
- **Information loss** when a model is used to represent reality.
- Less information loss → better model
Lower AIC → better predictive performance

- **AIC Formula**

$$AIC = 2k - 2\ln(L)$$

- Where:
- k = number of model parameters
- L = maximum likelihood of the model

Information Criteria-Based Model Selection

Akaike Information Criterion (AIC)

AIC Formula

$$AIC = 2k - 2\ln(L)$$

- Where:
- k = number of model parameters
- L = maximum likelihood of the model
- **Term 1: $2k$ — Complexity Penalty**
- **What does it mean?**
- Each parameter:
- Gives the model **more flexibility**
- Increases risk of **memorizing noise**
- So AIC **penalizes every extra parameter.**
- **Why the factor 2?**
- It comes from Akaike's theoretical derivation using:
- Kullback–Leibler divergence
- Asymptotic approximations
- More parameters $\rightarrow$ higher penalty $\rightarrow$ higher AIC

Term 2: $-2\ln(L)$ —Lack of Fit What is likelihood?

Likelihood measures:

How probable the observed data is under the model.

- Small errors $\rightarrow$ high likelihood
- Large errors $\rightarrow$ low likelihood

Taking:

$$-2\ln(L)$$

converts this into a **positive error-like quantity.**

Information Criteria–Based Model Selection

Akaike Information Criterion (AIC)

Connection to Squared Error (Very Important)

- For regression with Gaussian noise:

$$-2\ln(L) \propto \sum (y_i - \hat{y}_i)^2$$

- Minimizing squared error
Maximizing likelihood
Minimizing AIC
- → **All are connected**
- **How AIC Creates Balance**
- AIC balances **two competing forces**:
- **1. Fit Improvement**
 - Adding parameters reduces error
 - Increases likelihood
 - Decreases $-2\ln(L)$
- **2. Complexity Cost**
 - Adding parameters increases $2k$
 - Penalizes flexibility

Why AIC Has a Minimum

Initially:

- Adding parameters improves fit a lot

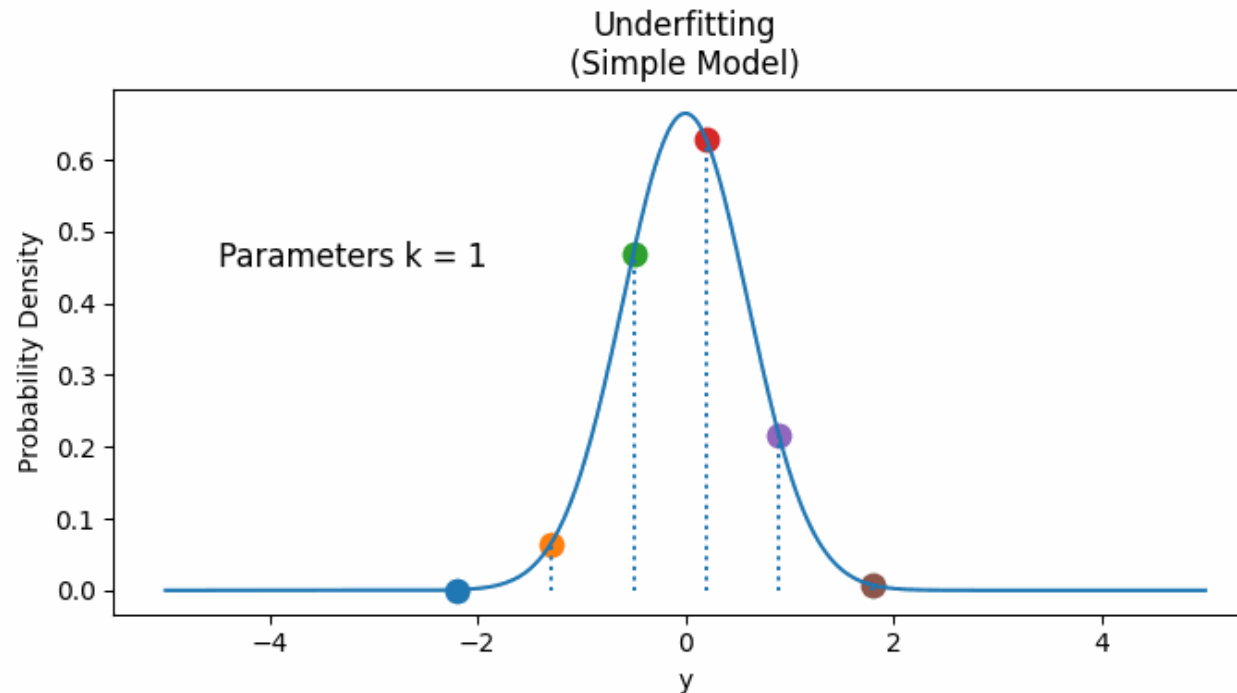
Later:

- Fit improves very little
- Penalty dominates
 - AIC decreases, then increases
 - Minimum AIC gives **best trade-off**

Information Criteria-Based Model Selection

Akaike Information Criterion (AIC)

Model Type	Fit	Penalty	AIC
Too simple	Poor	Small	High
Too complex	Very good	Large	High
Balanced	Good	Moderate	Low



Information Criteria–Based Model Selection

- **Bayesian Information Criterion (BIC)**

Why BIC Was Introduced

- BIC arises from **Bayesian probability theory**.
- It answers a **different question** from AIC:
- *Which model is most likely to be the true data-generating model?*

- **BIC Formula**

$$\text{BIC} = k \ln(n) - 2 \ln(L)$$

- Where:
- n = number of data points

Information Criteria–Based Model Selection

Bayesian Information Criterion (BIC)

- **Why Does BIC Penalize More?**

- As dataset size grows:

$$\ln(n) > 2$$

- So:
- Each extra parameter costs more
- Complexity becomes expensive
- Simpler models are favored

BIC becomes stricter with more data

Information Criteria–Based Model Selection

Philosophical Difference

Aspect	AIC	BIC
Theory	Information theory	Bayesian inference
Goal	Best prediction	True model
Penalty	Fixed (2k)	Grows with data
Behavior	Allows complexity	Prefers simplicity

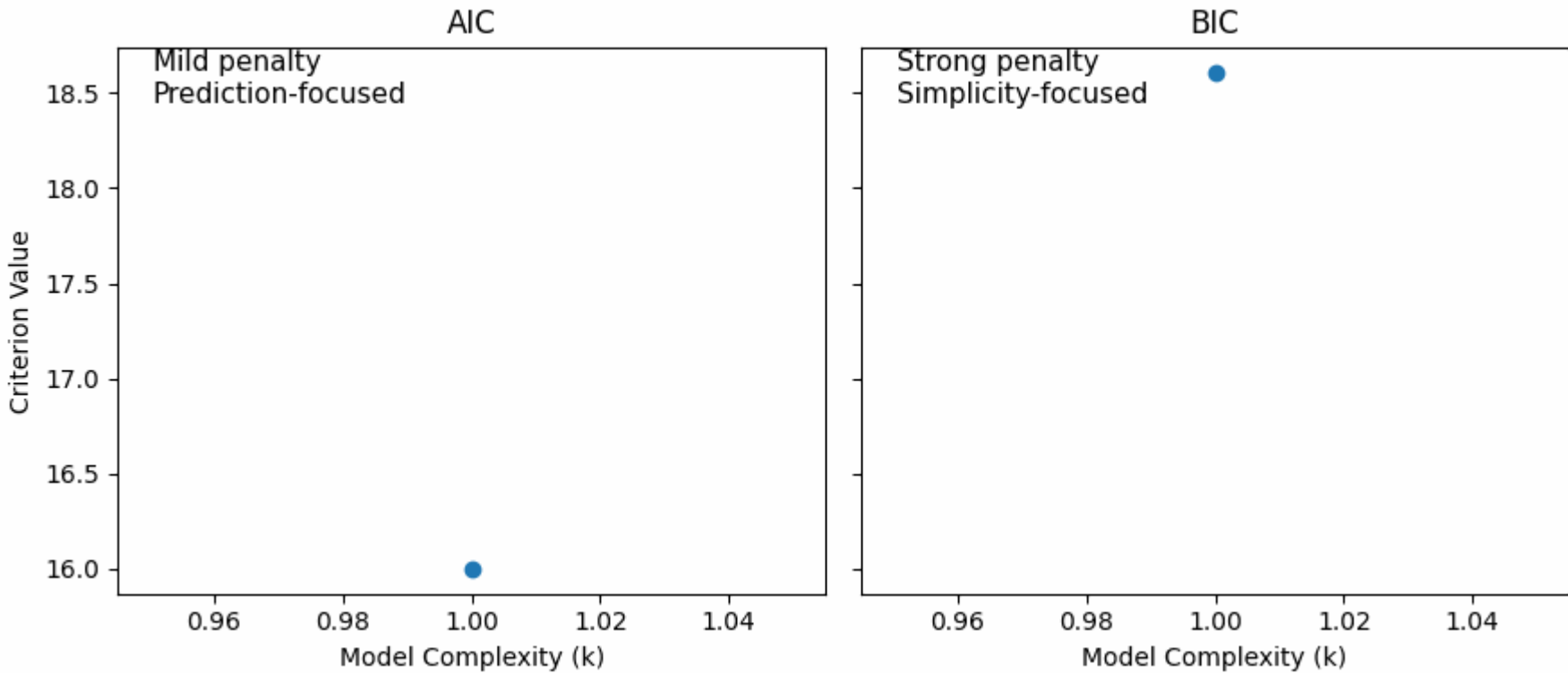
Weather Prediction

- AIC:
“Which model predicts tomorrow best?”
- BIC:
“Which model best explains the climate system?”
- **When AIC and BIC Disagree**
- Small dataset → similar choice
- Large dataset → BIC chooses simpler model
- AIC may choose complex model for prediction
- **Both are correct – different goals**

Information Criteria-Based Model Selection

AIC vs BIC

Criterion	Focus
AIC	Prediction accuracy
BIC	Model simplicity & truth



Information Criteria–Based Model Selection

- **Problem Statement**

- A data scientist fits three regression models to predict **electricity consumption**.

Model	Description	Parameters (k)	SSE
M_1	Linear	2	300
M_2	Polynomial (deg 2)	3	220
M_3	Polynomial (deg 5)	6	190

Number of samples:

$$n = 120$$

Information Criteria–Based Model Selection

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M_1	Linear	2	300
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M_3	Polynomial (deg 5)	6	190

Step 1: Compute Log-Likelihood

$$\ln(L) = -\frac{n}{2} \ln(SSE)$$

Model M_1

$$\ln(L) = -60 \ln(300) = -342.3$$

Model M_2

$$\ln(L) = -60 \ln(220) = -323.5$$

Model M_3

$$\ln(L) = -60 \ln(190) = -314.6$$

Information Criteria–Based Model Selection

- **Problem Statement**
- A data scientist fits three regression models to predict **electricity consumption**.

Model	Description	Parameters (k)	SSE
M ₁	Linear	2	300
M ₂	Polynomial (deg 2)	3	220
M ₃	Polynomial (deg 5)	6	190

Step 2: Compute AIC

$$AIC = 2k - 2\ln(L)$$

Best AIC → M₃

Model	AIC
M ₁	(4 + 684.6 = 688.6)
M ₂	(6 + 647 = 653)
M ₃	(12 + 629.2 = 641.2)

Information Criteria–Based Model Selection

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- A data scientist fits three regression models to predict **electricity consumption**.

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M_1	Linear	2	300
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Step 3: Compute BIC

$$\text{BIC} = k\ln(120) - 2\ln(L)$$
$$\ln(120) \approx 4.79$$

Best BIC $\rightarrow M_3$

Model	BIC
M_1	$(2(4.79) + 684.6 = 694.2)$
M_2	$(3(4.79) + 647 = 661.4)$
M_3	$(6(4.79) + 629.2 = 657.9)$

Information Criteria–Based Model Selection

- **Problem Statement**

- A data scientist fits three regression models to predict **electricity consumption**.

Model	Description	Parameters (k)	SSE
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M_2	Polynomial (deg 2)	3	220
M_3	Polynomial (deg 5)	6	190

- AIC allows complexity $\rightarrow M_3$
- BIC favors simplicity $\rightarrow M_1$

Information Criteria–Based Model Selection

Problem Statement

- A data analyst fits **two regression models** to a dataset with **n = 150 observations**.

Model	Description	Parameters (k)	SSE
M ₁	Simple Linear	2	250
M ₂	High-degree Polynomial	8	210

$$AIC = n \ln\left(\frac{SSE}{n}\right) + 2k$$

$$BIC = n \ln\left(\frac{SSE}{n}\right) + k \ln(n)$$

$$\ln(150) \approx 5.01$$

Information Criteria–Based Model Selection

Problem Statement

- A data analyst fits **two regression models** to a dataset with **n = 150 observations**.

Model	Description	Parameters (k)	SSE
M ₁	Simple Linear	2	250
M ₂	High-degree Polynomial	8	210

AIC Calculation Model M₁

$$\begin{aligned}\frac{250}{150} &= 1.667 \\ \text{AIC}_1 &= 150\ln(1.667) + 2(2) \\ &= 150(0.511) + 4 \\ &= \boxed{80.65}\end{aligned}$$

Model M₂

$$\begin{aligned}\frac{210}{150} &= 1.40 \\ \text{AIC}_2 &= 150\ln(1.40) + 2(8) \\ &= 150(0.336) + 16 \\ &= \boxed{66.40}\end{aligned}$$

AIC selects Model M₂

- Focuses on **predictive accuracy**
- Accepts higher complexity if fit improves

Information Criteria–Based Model Selection

Problem Statement

- A data analyst fits **two regression models** to a dataset with **n = 150 observations**.

Model	Description	Parameters (k)	SSE
M ₁	Simple Linear	2	250
M ₂	High-degree Polynomial	8	210

BIC Calculation Model M₁

$$\begin{aligned}
 \text{BIC}_1 &= 150\ln(1.667) + 2(5.01) \\
 &= 76.65 + 10.02 \\
 &= \boxed{86.67}
 \end{aligned}$$

Model M₂

$$\begin{aligned}
 \text{BIC}_2 &= 150\ln(1.40) + 8(5.01) \\
 &= 50.40 + 40.08 \\
 &= \boxed{90.48}
 \end{aligned}$$

BIC selects Model M₁

- Penalizes complexity more strongly
- Prefers **simpler model** when improvement is not overwhelming

Information Criteria–Based Model Selection

How to Interpret Information Criteria

Important Rules

- Compare AIC/BIC **only on same dataset**
- Absolute value does not matter
- Only **difference matters**

Δ AIC Rule of Thumb

Δ AIC	Interpretation
< 2	Models are similar
4–7	Moderate difference
> 10	Strong evidence